

MODULE 1 ASSIGNMENT

Course: PCCST501 – Computer Networks

Module: 1

Total Marks: 15

Submission: GitHub Classroom

Due Date: 20-06-2026

Assignment Title:

"How the Internet Delivers a Web Page"

Problem Statement

Choose **one website** that you frequently use (e.g., Google, Amazon, YouTube, Gmail, Instagram, or your college website). Explain the complete process that occurs when a user types the website URL into a web browser and presses **Enter**.

Your report should include:

1. Draw the TCP/IP protocol stack and explain the role of each layer.
2. Explain the role of **DNS** in converting the domain name into an IP address.
3. Describe how **HTTP** is used to request and receive the webpage.
4. Explain how **TCP** provides reliable communication for HTTP.
5. Identify whether the application follows a **Client–Server** or **Peer-to-Peer** architecture and justify your answer.
6. Compare **HTTP** and **FTP** with at least four differences.
7. Explain how **cookies** improve user experience with one real-world example.
8. Draw a flow diagram showing the complete communication from the browser to the web server and back.
9. Write a brief conclusion (100–150 words) describing the importance of application-layer protocols in everyday Internet usage.

Submission Guidelines

- **Length:** 3–5 pages
- **Format:** PDF
- **Diagrams:** Hand-drawn or created using any diagram tool